COURSE PRJECT REPORT
ON

RFID BASED ATTENDANCE SYSTEM

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1. INTRODUCTION

1.1. *Literature Survey*

1.1.1 Introduction

In recent years, security and access control have undergone significant advancements with the integration of modern technologies such as Radio Frequency Identification (RFID), biometric authentication, and wireless communication. Traditional mechanical lock-and-key systems have proven vulnerable to unauthorized access, duplication, and operational inefficiencies. To overcome these limitations, researchers have explored RFID-based door lock systems that combine security, convenience, and automation. This literature review discusses three major studies that focus on RFID-based access control and enhanced authentication mechanisms, highlighting their working principles, technological integration, and contributions to the development of intelligent and reliable security systems.

1.1.2 Description

Recent studies in this field highlight the increasing relevance of RFID door lock systems in modern security infrastructures. These systems typically comprise RFID tags, readers, controllers, and electronic locking units, which collectively facilitate rapid and secure authentication of users. RFID technology has demonstrated significant advantages over conventional access methods by enabling contactless operation, real-time data transfer, and remote monitoring through mobile and IoT integration. Researchers have also explored the use of Bluetooth communication to relay door status updates directly to users, thereby improving accessibility and situational awareness.

The literature emphasizes that RFID systems are not only beneficial for residential and commercial applications, but also for institutional and industrial environments where multiple levels of access control are required. These systems provide speed, reliability, and scalability, allowing them to be easily adapted for diverse operational conditions. Furthermore, studies have examined advanced versions that combine RFID with biometric fingerprint sensors and Global System for Mobile (GSM) modules, creating a three-factor authentication framework. This integration ensures multi-layered security by validating a user's RFID tag, fingerprint, and a unique token sent to the registered mobile device. Performance evaluations of such systems have shown over 98% accuracy in distinguishing authorized users from unauthorized ones, underscoring their reliability and robustness.

Despite these advantages, challenges persist in the form of data security vulnerabilities, signal interference, and high installation costs. Research suggests that incorporating encryption mechanisms, signal filtering, and cost-effective hardware solutions can mitigate these issues. The combination of RFID with IoT and mobile technologies has been recognized as a promising approach for enhancing system intelligence, enabling remote access control, automated monitoring, and user data management. Overall, the integration of RFID technology with emerging digital platforms continues to transform traditional access systems into intelligent, secure, and user-friendly solutions.

1.1.3. Conclusions

From the reviewed literature, it is evident that RFID-based access control systems represent a major technological advancement in modern security design. The combination of RFID with IoT, Bluetooth, GSM, and biometric technologies has paved the way for creating multi-factor, automated, and highly secure locking mechanisms. These systems address the key limitations of conventional locks by

improving accuracy, efficiency, and scalability. While challenges such as cost and interference remain, ongoing research and innovation are steadily overcoming them. The continued development of RFID-integrated access systems signifies a move towards smart, reliable, and adaptive security solutions suitable for residential, commercial, and industrial environments.

1.1.4 References

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1.2. Aim

The aim of this project is to design and implement an RFID-based attendance system that automates the process of recording and managing attendance using Radio Frequency Identification (RFID) technology. The system seeks to ensure accurate, efficient, and secure identification of individuals by replacing traditional manual attendance methods with a contactless and real-time solution, thereby minimizing errors, preventing proxy attendance, and improving data management in educational and organizational environments.

1.3. Objectives

- Study the working of RFID: Understanding how electromagnetic induction and radio waves enable wireless communications.
- Implementing RFID as a sensor: Using RFID tags to identify and detect objects without physical contact.
- Analysing the applications: To analyse the real-world applications, such as security access, tracking and inventory management.
- Evaluating Feasibility: Evaluating pros and cons, accuracy, cost and efficiency.

2. HARDWARE COMPONENTS USED

1. RFID reader module
2. RFID tags
3. Microcontroller (Arduino/ESP32)
4. Connecting wires
5. Breadboard

6. Servo motor
7. Buzzer
8. LCD
9. LEDs

3. WORK DONE

3.1 Implementation of LCD:

The I2C Liquid Crystal Display (LCD) was implemented as the primary user interface for real-time feedback.

- Library and Initialization: The LiquidCrystal_I2C"library was used to simplify communication. The LCD was initialized in setup() with its specific I2C address (0x27) and dimensions (16x2). The backlight was also activated for visibility.
- Centralized Display Function: To standardize all screen updates and prevent code repetition, a single function named updateDisplay(String str) was created. This function clears the screen, sets the cursor to the top-left corner, and prints the new string.
- Role in the System: The LCD is used to display all system states to the user, including:
    - Idle State: "RFID Attendance" (shown on boot and after a 5-second inactivity timeout).
    - User Data: Displays the full name of a user immediately after their card is scanned.
    - List Browsing: Shows individual names from the names array as the user scrolls with the buttons.
    - Status/Error Messages: Provides feedback like "No records found," "Invalid Card," or "List Uploaded."

3.2. Implementation of MFRC522 (RFID)

The MFRC522 RFID module, communicating over the SPI bus, is the project's core data input device. Its implementation handles card detection, authentication, and complex data parsing.

- Initialization: The MFRC522 library was included, and an instance was created by defining its SS (pin 5) and RST (pin 4) pins. The SPI bus (SPI.begin()) and the RFID module (mfrc522.PCD_Init()) were initialized in setup().
- Polling and Detection: The main loop() constantly polls for a new card using mfrc522.PICC_IsNewCardPresent(). When a card is detected, it calls the custom addName() function to handle the entire read process.
- Custom Data Reading Functions: Reading a full name stored across multiple blocks required two custom functions:
    1. rfid_get(byte block, byte *buffer): This is a low-level function that handles reading a single 16-byte block. It authenticates the block using the default MIFARE key (0xFF...) and then uses mfrc522.MIFARE_Read() to pull the data into a buffer.

2. rfid_read(byte blockA, byte blockB): This is a high-level function that reads a 32-byte name. It calls rfid_get() twice (once for blockA and once for blockB) to get the first and last name data.

- Data Processing and Cleaning: The 32 bytes of raw data read from the card are not a clean string.
  - The two 16-byte buffers are combined into a single 32-byte nameBuffer.
  - The code then iterates through this buffer and filters the data, adding only printable ASCII characters (values 32 to 126) to a new String variable.
  - Finally, name.trim() is used. This is a critical step that removes all the extra padding spaces (which were added by the write sketch) and any other invisible whitespace from the beginning or end of the string.
- Data Storage: The resulting clean String (e.g., "Jacob Chacko") is added to the names[10] array, and the nameCount variable is incremented, effectively "registering" the user.

3.3 Implementation of Bluetooth Serial

The BluetoothSerial library was used to wirelessly export the list of scanned names from the ESP32 to a connected device, such as a mobile phone.

- Initialization: The library was included, and the Bluetooth controller was initialized in setup() with a public device name: SerialBT.begin("RFID Attendance"). This makes the ESP32 discoverable by a phone.
- Trigger-Based Transmission: To save power and not constantly transmit, the Bluetooth function is triggered by a specific user action: pressing both buttons simultaneously.
- Dual-Press and Debounce Logic:
  - The code checks if both buttons are *held* (a_isHeld && b_isHeld).
  - To prevent the list from being sent thousands of times per second, two layers of control were added:
    1. State Change Detection: The (!btnA_wasHeld || !btnB_wasHeld) logic ensures the code fires only *once* when the buttons are first pressed down.
    2. Cooldown Timer: A lastSendTime variable enforces a 2-second (2000ms) cooldown, preventing the user from accidentally sending the list multiple times by holding the buttons down.

Data Transmission: When triggered, the system iterates through the names array from 0 to nameCount and sends each name using SerialBT.println(). The .println() function is used so that each name appears on a new line in the phone's Bluetooth serial terminal app, resulting in a clean, readable list. The LCD is then updated with "List Uploaded" as confirmation.

4. SOFTWARE CODING

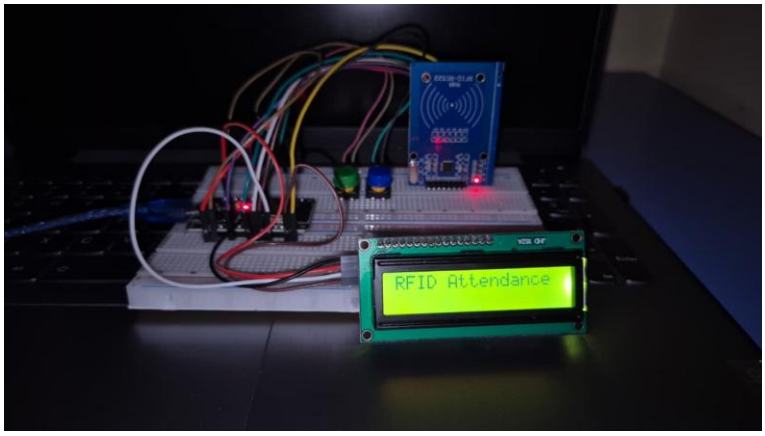
rfid_attendance_v1.ino

```
#include <SPI.h>
#include <MFRC522.h>
#include <LiquidCrystal_I2C.h>
#include <Button.h>
#include <BluetoothSerial.h>

#if !defined(CONFIG_BT_ENABLED) || !defined(CONFIG_BLUEDROID_ENABLED)
#error Bluetooth is not enabled! Please run `make menuconfig` to and enable it
#endif

#define RST_PIN         4
#define SS_PIN          5

BluetoothSerial SerialBT;

MFRC522 mfrc522(SS_PIN, RST_PIN);   // Create MFRC522 instance.
MFRC522::StatusCode status;

LiquidCrystal_I2C lcd(0x27, 16, 2);

String names[10];
int nameIndex = 0;

int nameCount = 0;

Button btnA(25); //back
Button btnB(26); //next
bool btnA_wasHeld = false;
bool btnB_wasHeld = false;
unsigned long lastSendTime = 0; // Tracks when we last sent the list
unsigned long lastAct = 0;

void setup() {
  Serial.begin(9600);
  SPI.begin();        //RFID
  mfrc522.PCD_Init();

  lcd.init();       //LCD
  lcd.backlight();

  btnA.begin();     //buttons
  btnB.begin();

  SerialBT.begin("RFID Attendance"); //Bluetooth device name

  updateDisplay("RFID Attendance");
}

void loop() {
  unsigned long time = millis(); // Get the current time

  bool a_isHeld = (btnA.read() == Button::PRESSED);
  bool b_isHeld = (btnB.read() == Button::PRESSED);

  if (btnA.pressed()) {
    if (nameCount==0) {
      updateDisplay("No records found");
    } else {
      nameIndex--; // Go to the previous name
      if (nameIndex < 0) {      //bounds checking
        nameIndex = nameCount - 1;
      }
      updateDisplay(names[nameIndex]);
    }
    lastAct=time;
  }

  if (btnB.pressed()) {
    if (nameCount==0) {
      updateDisplay("No records found");
    } else {
      nameIndex++; // Go to the next name
      if (nameIndex >= nameCount) {   //bounds checking
        nameIndex = 0;
      }
      updateDisplay(names[nameIndex]);
    }
    lastAct=time;
  }

  if (a_isHeld && b_isHeld && (!btnA_wasHeld || !btnB_wasHeld)) {
    if (time - lastSendTime > 2000) {
      for (int i = 0; i < nameCount; i++) {
        SerialBT.println(names[i]);
      }
      updateDisplay("List Uploaded");
      lastAct=time;
      lastSendTime = time;
    }
  }

  MFRC522::MIFARE_Key key;
  for (byte i = 0; i < 6; i++) key.keyByte[i] = 0xFF;

  if ( mfrc522.PICC_IsNewCardPresent()) {
    addName();
    lastAct=time;
  }

  if (time - lastAct > 5000) {
    updateDisplay("RFID Attendance");
    lastAct = time;
    nameIndex= nameCount-1;
```

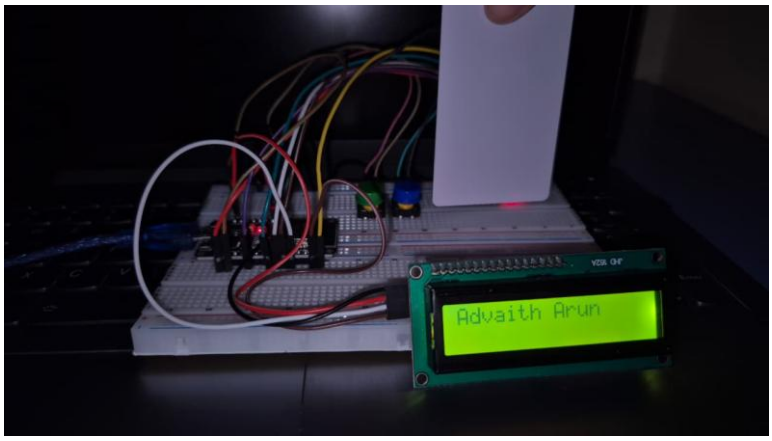
```
  }
}

void rfid_get(byte block, byte *buffer) {
  MFRC522::MIFARE_Key key;
  for (byte i = 0; i < 6; i++) key.keyByte[i] = 0xFF;

  MFRC522::StatusCode status;
  status = mfrc522.PCD_Authenticate(MFRC522::PICC_CMD_MF_AUTH_KEY_A, block, &key, &(mfrc522.uid));
  if (status != MFRC522::STATUS_OK) {
    Serial.print("PCD_Authenticate() failed: ");
    Serial.println(mfrc522.GetStatusCodeName(status));
    return;
  }

  byte len = 18;
  status = mfrc522.MIFARE_Read(block, buffer, &len);
  if (status != MFRC522::STATUS_OK) {
    Serial.print("Reading failed: ");
    Serial.println(mfrc522.GetStatusCodeName(status));
    return;
  }
}

String rfid_read(byte blockA, byte blockB) {
  byte buffer1[18]; byte buffer2[18];
  rfid_get(blockA,buffer1); rfid_get(blockB,buffer2);
  byte nameBuffer[34];
  String name;

  for (byte i = 0; i < 16; i++) {
    nameBuffer[i] = buffer1[i];
  }
  for (byte i = 0; i < 16; i++) {
    nameBuffer[i+16] = buffer2[i];
  }

  for (uint8_t i = 0; i < 30; i++)
  {
    if (nameBuffer[i] >= 32 && nameBuffer[i] <= 126)
    {
      name += (char)nameBuffer[i];
    }
  }
  name.trim();
  return name;
}

void addName() {
  MFRC522::MIFARE_Key key;
  for (byte i = 0; i < 6; i++) key.keyByte[i] = 0xFF;

  if (!mfrc522.PICC_ReadCardSerial()) {
    updateDisplay("Invalid Card");
    return;
  }
  Serial.println("**Card Detected:**");

  String name;
  name = rfid_read(1,2) + " " + rfid_read(4,5);
  name.trim();

  mfrc522.PICC_HaltA();
  mfrc522.PCD_StopCrypto1();

  if (name.length() == 0) {
    updateDisplay("Invalid Card");
    return;
  } else {
    updateDisplay(name);
    names[nameCount]=name; nameCount++;
    return;
  }
}

void updateDisplay(String str) {
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print(str);
  Serial.println(str);
}
```

5. RESULTS & DISCUSSION

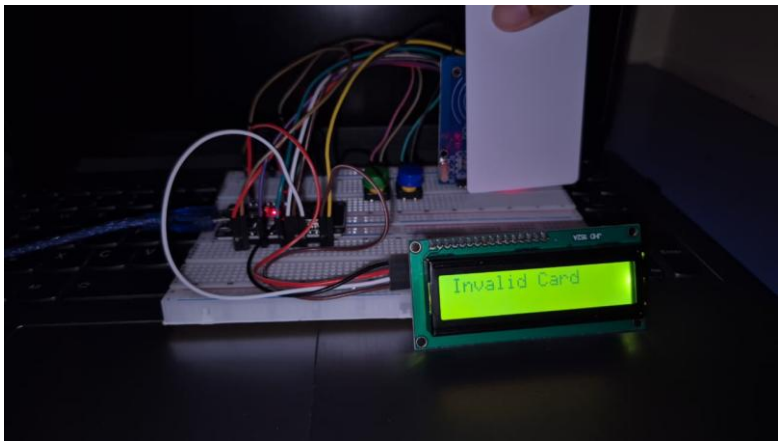
1. Idle State:



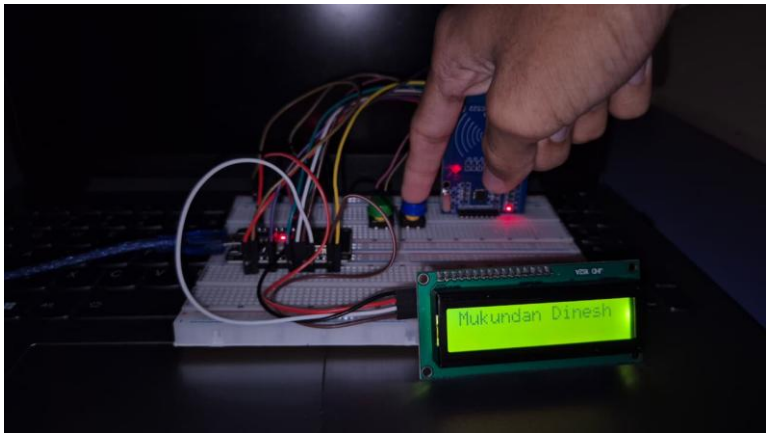
2. Card Scanning



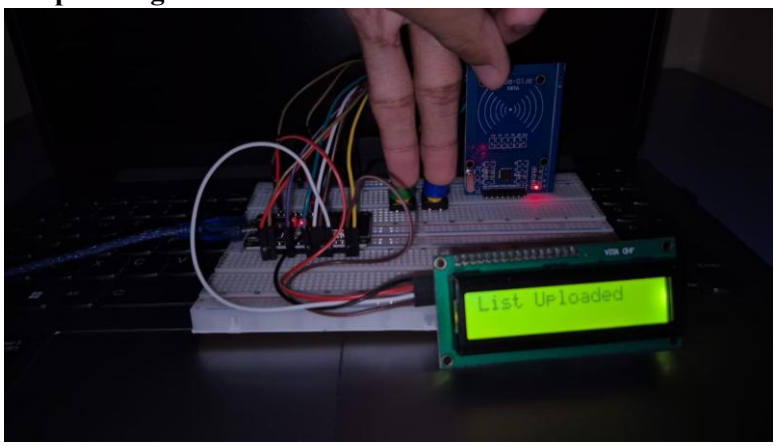
3. Invalid Entry:



4. Previous Entries



5. Uploading list to device



6. Uploaded List

6. FUTURE WORK

- **Scalable Centralized Database**: The system can be upgraded with a scalable centralized database to handle a larger number of users and records efficiently. This would improve performance, enable real-time data synchronization, and make the system suitable for use across multiple departments or locations.
- **Durable and Aesthetic Enclosure**: The redesign of the hardware enclosure is for durability and better appearance; the use of stronger materials and compactness in design will protect the components from damage, giving it a professional look and making the device appear modern.
- **Enhanced RFID Card Security**: RFID cards could be made more secure using encryption or authentication techniques that prevent unauthorized access or data duplication. The security would be further improved in systems where RFID identification is combined with other methods of verification.

7. REFERNCES

- A. Nair, R. Adithyan, A. Unni and S. Nalinakshan, "RFID Door Lock Access Control Systems: Trends, Technologies and Applications," *2025 3rd International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT)*, Bengaluru, India, 2025, pp. 906-912, doi: 10.1109/IDCIOT64235.2025.10914928.
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